

Marmara University

Engineering Faculty



Electrical and Electronics Engineering

TRAINING REPORT

Student

Name: Melih Alparslan Aksoy

Student ID: 150717003

The training is for

☒ EE300

☐ EE400

microsonic GmbH

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30.08.2024



Electrical and Electronics



Engineering Department

TRAINING REPORT

Student

Name : Melih Alparslan Aksoy

Student ID : 150717003

**Department/semester : Electrical & Electronics
Engineering**

The company name : Microsonic GmbH

Company Department	Work Done	DATE		Department Head
		Start	End	
Embedded Development	Embedded Communication	04.07.24	28.08.24	Muhammed Tatlısu

Stamp (of the Company) & Signature (Executive Officer)

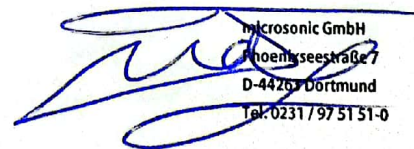
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30.08.2024



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Weekly Schedules: 1st Week

Date	Tasks Accomplished
01/07/2024	
02/07/2024	
03/07/2024	
04/07/2024	This is my first day in the company and I met with other colleagues. Then, I did some research about how to work with microcontrollers and its pins. Then, in the company they allowed me to take a test about ultrasonic distance sensor whether it works on different degrees like 0,40 or 60 Celsius etc.
05/07/2024	Today we continued to make the remaining test degrees. I gathered the curve views and their values inside files to compare them and get a result for the unwanted conditions.
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Weekly Schedules: 2nd Week

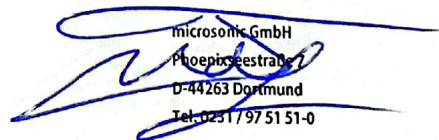
Date	Tasks Accomplished
08/07/2024	I continued my mission for toggling the LED on microprocessor. It was the first project of mine. I continued to try and make a result but today was not enough so I will continue tomorrow.
09/07/2024	Today we completed the last week experiment. I made some test for 2 new ultrasonic sensors to see if the changes of sensor, but same material affects anything.
10/07/2024	At this time, I made it to turn on and off one of the LED on microprocessor. I took as an example my Monday knowledge and added viewpoint while checking the Project Explorer tab of Code Composer Studio.
11/07/2024	Then my supervisor gave me another mission to toggle the light of microcontroller's while pressing an external button. The Button was attached a white board, and I connected it to one of the Pins of the microcontroller.
12/07/2024	Today I made to achieve the desired mission. I programmed the software that whenever a button has been clicked, it should toggle.
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Weekly Schedules: 3rd Week

Date	Tasks Accomplished
15/07/2024	Today I got another mission. It works like whenever a button on the microcontroller pressed now the LED must toggle. I got the brief and started to work on it.
16/07/2024	This day I completed to toggle whenever button has been clicked. So that this was the first Communication experiment. It was for one Microcontroller, but I handled it.
17/07/2024	Today I started a new project. Whenever a button on one MCU, the LED on other MCU's must be toggled and reversely. So, it is a new topic for me because I always working on just one MCU.
18/07/2024	I started to learn more about the other microcontroller. How it works how it starts and so on. They were so similar but different hardware, so it was easy for me to understand that.
19/07/2024	I completed project. I did some research in the morning the connect the pins and get a new sight of view, so it made my mind broadener.
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Weekly Schedules: 4th Week

Date	Tasks Accomplished
22/07/2024	Today it was another meeting to what to do after previous project. I started to communicate two Microcontrollers. When one sends a byte to another, the other MCU must send a response if it is desired situation or not. So, I started to work on it.
23/07/2024	I continued to work on the same project. I collected more information on web and on Code Composer Studio itself. I did some tries and saw the logic.
24/07/2024	I made it. I completed the software and hardware. I was able to reach the goal. One controller sends a byte and other returns a response.
25/07/2024	Today I got brief for another project. It is similar but more difficult because now one microcontroller sends 4 bytes of data, the other send another response. When the response is desired first microcontroller sends another byte, and the second one should send another response byte too.
26/07/2024	So, I completed. It was hard for yesterday but today I made various tries and got the result. Whenever 4 bytes is sent other says okay or not. When the result is Okay, first MCU send another 4 bytes.
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Weekly Schedules: 5th Week

Date	Tasks Accomplished
29/07/2024	I have been briefed that I will use communication method with different approach so that the data package will be different as an unusual situation. It's rate and speed must not affect the data response of responding MCU.
30/07/2024	Today I clarified which method I should use. I searched and grabbed the information about the project, its meaning and logic.
31/07/2024	I continued my work on Communication with both Microcontrollers. I made various attempts. I learnt my mistakes and make more approaches.
01/08/2024	I learnt how to XOR bridge works to determine if the transmitted and received data are same and equal or not. So with using that meaning, we can define if it's valid or not.
02/08/2024	Today all day I was working to pass data from one device to another using Checksum method. It was tough to find the method but at the end I made it.
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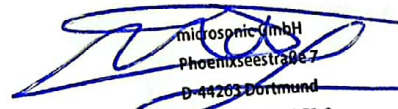
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Weekly Schedules: 6th Week

Date	Tasks Accomplished
05/08/2024	The communication was not complete and made it completed today. I put some additional commands to make it as a whole and it runs well.
06/08/2024	Today, I started to make my code as dynamic so that it works in any conditions. I started to implement some values and variables and made some logical methods so it adapts every situation.
07/08/2024	This day I had a meeting and got another mission so that communication should be worked as mutually. That means when one sends a data to other, second board should response and after that response the first controller should send another data to approve.
08/08/2024	I continued to make the algorithm. I made some changes and research to how to make it and made lots of tries. I eventually made progress and hopefully will solve it on Friday.
09/08/2024	This day I completed the code. It works in good conditions, and it is also dynamic. So that it works in every condition in terms of changing data size, interrupts etc.
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Weekly Schedules: 7th Week

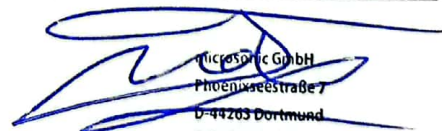
Date	Tasks Accomplished
12/08/2024	Today there was one mistake in the code. I solved it. That meant I completed the code, and I searched some new projects inside Project Explorer in Code Composer Studio to get new insight.
13/08/2024	I contacted with my supervisor and got another mission. He told me to use Ultrasonic Sensor (HC-SR04) so that I can measure the distance between any objects and the microcontroller up to some limitation.
14/08/2024	This day I made some research about HC-Sr04 and got information about how the Ultrasonic sensors work and how they measure the distance, emerge sound waves etc.
15/08/2024	I continued my research and found more data and projects, but they are on other microcontrollers of Texas Instruments, so I had to adapt logics behind them into my controller and I started coding.
16/08/2024	Today, I continued my coding and made an algorithm, but it did not go as I planned so I made some changes and kept going to understand which functions and methods I must use to integrate the circuit.
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Weekly Schedules: 8th Week

Date	Tasks Accomplished
19/08/2024	Today I made research about other Microcontrollers, how they work, how they react to the sensor measurement and so on. Because of their implementation methods are different then my previous, I had to cover and get the differences.
20/08/2024	I had a meeting with my supervisor about the current code and its operation. I corrected my code, and it gives results to screen but I guess it is not correct. I will use another approach.
21/08/2024	Today, I tried to fix the code, but it did not work either, so the problem was understood. It was the circuit. So, I made some changes and got a desired solution for the system.
22/08/2024	This day I managed to get a reasonable value from ultrasonic sensor but apparently my code won't work all day I tried to change it but hopefully on Friday I will make it work and measure the desired distance of any object into the range of sensor.
23/08/2024	Today I continued my experiment and nearly got some results. I improved the code and hardware but still got some problems because of moving the objects which should be measured by the sensor. I will continue to finish the project next week with bigger enthusiasm.
Executive Officer Muhammed Tathisu	

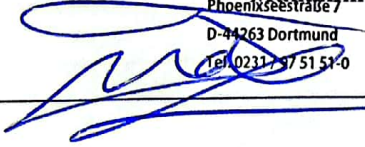

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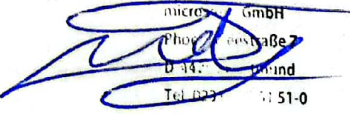
Weekly Schedules: 9th Week

Date	Tasks Accomplished
26/08/2024	Today, I managed to measure the distance with using ultrasonic sensor. I improved my code and made the success. After that I tried to make another logic as when the distance is less then 15 cm, it should turn on green LED from the whiteboard or yellow LED oppositely. The result and which LED should turn on is decided by second microcontroller so communication between microcontroller took role here.
27/08/2024	Today, I studied on how to get a security range if the measurement changes compared to previous measurement and current one. Then improved yesterday's LED circuit with that logic. Then I got a Led display screen to print the measurement. I configured pins, made the circuit with transistors and got some results but it's not completely correct so I will make better tomorrow.
28/08/2024	This day, I continued to finish the project. I enabled Led display screen and configured inside Code Composer Studio with using C language. The goal was to display the distance which was coming from first board. I made it and got the results after every measurement. So the project has been completed.
29/08/2024	-
30/08/2024	-
Executive Officer  microsonic GmbH Phoenixseestraße 7 D-44263 Dortmund Tel: 02331 97 51 51-0 Muhammed Tatlısu	



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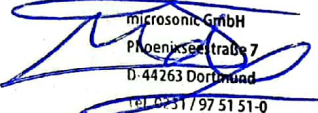
1. INTRODUCTION

The Embedded Programming is in everyday life. It can be in anywhere where you see electronics all over world. You can see some boards under remote controllers of air conditioners, televisions or under the battery of your mobile phone. It can be seen better in if a old type phone like not-smart ones. Don't open your current phone's back cover to see if there is a board or not. You can break it. No liability accepted.

When mentioning about Embedded Programming and its using areas, it's software and hardware are so important to make the system work in good and desired conditions. Therefore, it can be considered as a whole. When a developer improves a software, he/she should also consider the hardware part to integrate everything. When it is thought that a project is so simple and can be applied to quick, it is the wrong sight of thinker because of wrong calculation of enabling software, hardware or both. So as I said, it must be considered as a whole.

In this project C language has been used. C is a high developed and standardised programming language. C was firstly related with assembly languages and its applications but in the developed and renewed versions is nearly adopted in every area of programming world. New versions like C++ or C# can be seen in some applications. They have all their own advantages and disadvantages so user/programmer might decide in which lean the software go and use that language.

I myself took STEM lecture in Marmara University. That means I got some experiences from previous semesters. I firstly took C language lesson in the first semester of my university life then C++, STEM embedded, Matlab, Java and Python came up. When the logic behind coding understood, other languages may not be hard to cover so first touch to Software Languages is so crucial.

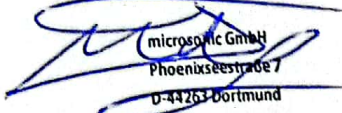

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2. ABOUT THE COMPANY

Microsonic GmbH is a company based in Dortmund, Germany, known for its expertise in ultrasonic sensor technology. The company specializes in the development, production, and distribution of ultrasonic sensors for various industrial applications. These sensors are used for distance measurement, object detection, proximity sensing in.

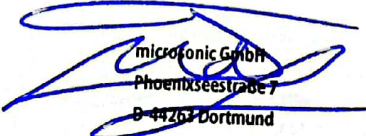


(Company building is in Phoenixseestrasse 7,44263,Dortmund/Germany)

Sensors of the company are used in various areas in different locations of working environment. They are used in Automation Technology, Robotics, Agriculture, Logistic, Manufacturing, Packaging. The company itself is an idea to give more value into innovation, manufacturing and creating new systems and sensors to upgrade.

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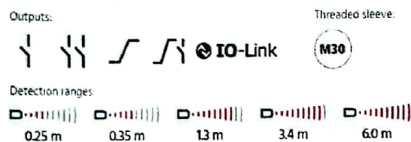

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There are lots of sensor types in the company. For giving an example, I could give this distance sensor example. The mic+ sensor family features ultrasonic sensors in a compact M30 housing, capable of measuring distances from 30 mm to 8 m with high resolution. These sensors offer easy setup via TouchControl with an LED display, and multiple output options, including switching and analog. Advanced features include temperature compensation, synchronization of up to ten sensors, and a Teach-in process for quick calibration. The sensors can be configured through a PC using LinkControl and support IO-Link for enhanced communication in automation systems.

mic+ ultrasonic sensors

mic+ sensors are available in four unit variants with five different detection ranges.



[LEARN MORE](#)



(Image of mic+ sensor from microsonic.de web page)

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3. PRACTICAL TRAINING

I did my internship in Embedded Programming. I wanted to divide it into 3 different under topics: Software Development, Hardware Development and Sensor Experiment. These topics will be covered in upcoming sections. So, it will be explained more with related my project.

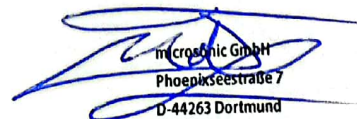
The main idea of this Internship Program is to make a view, understand the technique and work on communication between two different microcontrollers. I spent over 5 weeks to get to know the communication logic, how it works how it applies to different aims. The user may ask why it is so important to focus on communication between microcontrollers. The answer is simple. Let me give an example to illustrate. People have 2 arms. Let's think like he/she has to do 2 tasks, and these tasks need 2 hands at the same time but when another task comes up, person will challenge and may be not do it. Same is correct for microcontrollers. They have limited storages, pins, functions and capabilities to work in good conditions. If they help each other and share the job, it will bring more efficient, maybe time saving, more economical conditions. For this logic we wanted to work on communication of two microcontrollers. For the microcontrollers, they can share data between them, when one works on challenging and hard project, other can help it to make it simpler or when one works on one mission other can work on other mission so that they can gather some information at the same time and can use whatever they want. In Turkish there is a quote that "One hand has no sound, but two hands make noise."

3.1. Software Development

So, in this project I used Code Composer Studio (CCS) from Texas Instruments (TI) and C language to implement codes into Microcontrollers. The reason why I used C programming has been explained in above sections. Firstly, I tried to know and understand the Microcontrollers' working principles. Because of there were 2 different Microcontrollers, I tried to get to know if they have special features, functions, sensors etc. Then I realized that they are in the same family

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so there must be minimum differences between them especially in the meaning of coding and semantics.

There is a Project Explorer option in CCS. I used it to get a dive into some example projects. I got informed myself of how to work with Microcontrollers' special functions and references. At that time my C knowledge has been emerged. I searched more example projects and its mechanisms.

Create New Project

Microcontroller

Examples

adc_dma_ping_pong

adc_singlechannel

adc_to_i2c_target

adc_to_pwm

adc_to_spi_peripheral

adc_to_uart

adc12_14bit_resolution

adc12_internal_temp_sensor_mathacl

Out of Box experience for MSPM0 LaunchPads

Capture ADC data into 2 (ping-pong) buffers

Get the samples from the configured ADC channel and print them on the console using UART

Sample ADC and respond to I2C commands sent by a controller

Convert analog voltage to PWM

Sample ADC and respond to SPI commands sent by a controller

Configures the ADC0 to trigger single conversion and send the ADC data through UART

ADC12 conversion at maximum frequency using DMA with hardware averaging enabled

ADC12 samples internal temperature sensor and converts reading to Celsius and Fahrenheit degrees

(Here is an illustration of Project Explorer of Code Composer Studio.)

Inside every Project I opened, I took a look at original functions and how they piled up together.

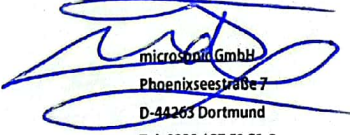
```

1 //
2 #include "ti_msp_dl_config.h"
3
4 int main(void)
5 {
6     SYS_CFG_DL_init();
7
8     DL_SYSCTL_enablesleepOnExit();
9
10    while (1) {
11        __WFI();
12    }
13 }

```

(This example code demonstrates how to publish and subscribe events between pins without CPU intervention.)

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After every code and iteration, I had to inform the microcontroller which pin I need to use so that I had to configure the pins in logical way. I learnt how to configure pins regularly as shown below.



(TIMER pin configuration example is given here.)

So now I knew how to use function, configuration portal and microcontroller behavior. Now I had to include the sensor and measure the distance every specific time interval. In that reason, I imported and added sensors pins to microcontroller. Ultrasonic sensor is named as HC-SR04 and it has its own characteristic pins. It has 4 pins: VCC (Initial Power pin), GND (Ground pin), TRIG (Triggered) und ECHO pins. The working principal is like that as follows. VCC pins takes 5 Volt from Microcontroller and a set value for TRIG pin to activate the sensor. Then sensor sends sound wave (40kHz) and get feedback from ECHO pin. Feedback is a reflection method. Sound wave emerged from TRIG pin reflects from a material whose distance is wanted to measure and ECHO pin captures it. Because of connecting pins between ECHO pin and

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microcontroller port, microcontroller reads the value of ECHO and measure its pulse width and measure the distance.

```
31
32 while (1) {
33     DL_Timer_stopCounter(CAPTURE_0_INST);
34     DL_TimerG_setTimerCount(CAPTURE_0_INST,0);
35     DL_TimerG_startCounter(CAPTURE_0_INST);
36     DL_GPIO_setPins(GPIO_PINS_PORT,GPIO_PINS_PIN_0_PIN);
37     Delay(33);
38     DL_GPIO_clearPins(GPIO_PINS_PORT,GPIO_PINS_PIN_0_PIN);
39     //DL_GPIO_togglePins(GPIO_LEDS_PORT,GPIO_LEDS_USER_LED_1_PIN);
40
41     while (false == gCheckEdgeCaptureTime) {
42         __NOP();
43     }
44     gCheckEdgeCaptureTime = false;
45     trx[0] = ((double)edgeCapture*10/58);
46     printf("value is %d cm\n",trx[0]);
47     DL_UART_transmitData(UART_0_INST,trx[0]);
48     while (!DL_UART_isTXFIFOEmpty(UART_0_INST));
49     //__BKPT(0);
50     Delay(60000);
51     //Delay(200000);
52     //break;"/
53 }
54
55
56 void CAPTURE_0_INST_IRQHandler(void)
57 {
58     switch (DL_TimerG_getPendingInterrupt(CAPTURE_0_INST)) {
59         case DL_TIMERG_IIDX_CC0_UP:
60             //turn=1;
61             edgeCapture = (DL_Timer_getCaptureCompareValue(CAPTURE_0_INST, DL_TIMER_CC_0_INDEX)); // - 2210;
62             DL_GPIO_togglePins(GPIO_LEDS_PORT,GPIO_LEDS_USER_LED_1_PIN);
63             gCheckEdgeCaptureTime=true;
64             break;
65         default:
66             break;
67     }
68     //default:
69     edgeCapture2 = (DL_Timer_getCaptureCompareValue(CAPTURE_0_INST, DL_TIMER_CC_0_INDEX));
```

(Here is the detailed code representation of Ultrasonic Sensor.)

After getting measurement values, I wanted to show them in LED display. To make it configuration and codes must be integrated. I read the LED display datasheet to learn about its pins and working principles. Then I created a function to print numbers from 0 to 9 on every segment of LED Display. Display has 3 different segments. So, I had to use another TIMER to use them whenever and however I wanted. Then It was the time to use UART module of these two microcontrollers to communicate them. I got two microcontrollers beginning of the projects. MSPM0G3507(red) and MSPM0C1104(green). Red microcontroller stores the distance measure

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value coming from ultrasonic sensor and green microcontroller gets every measurement and prints it on the LED Display.

```
void acq(uinsec_t nscr){
    switch (nscr){

        case 0:
            DL_GPIO_clearPins(GPIO_PINS_PORT,GPIO_PINS_PIN_6_PIN | GPIO_PINS_PIN_7_PIN | GPIO_PINS_PIN_9_PIN |GPIO_PINS_PIN_10_PIN | GPIO_PINS_PIN_8_PIN | GPIO_PINS_PIN_2_PIN);
            DL_GPIO_setPins(GPIO_PINS_PORT, GPIO_PINS_PIN_12_PIN);
            break;

        case 1:
            DL_GPIO_clearPins(GPIO_PINS_PORT,GPIO_PINS_PIN_8_PIN | GPIO_PINS_PIN_2_PIN);
            DL_GPIO_setPins(GPIO_PINS_PORT, GPIO_PINS_PIN_6_PIN | GPIO_PINS_PIN_7_PIN | GPIO_PINS_PIN_9_PIN |GPIO_PINS_PIN_10_PIN | GPIO_PINS_PIN_11_PIN);
            break;

        case 2:
            DL_GPIO_clearPins(GPIO_PINS_PORT, GPIO_PINS_PIN_6_PIN | GPIO_PINS_PIN_7_PIN | GPIO_PINS_PIN_8_PIN |GPIO_PINS_PIN_10_PIN | GPIO_PINS_PIN_11_PIN);
            DL_GPIO_setPins(GPIO_PINS_PORT,GPIO_PINS_PIN_9_PIN | GPIO_PINS_PIN_2_PIN);
            break;

        case 3:
            DL_GPIO_clearPins(GPIO_PINS_PORT, GPIO_PINS_PIN_6_PIN | GPIO_PINS_PIN_7_PIN | GPIO_PINS_PIN_8_PIN |GPIO_PINS_PIN_10_PIN | GPIO_PINS_PIN_11_PIN);
            DL_GPIO_setPins(GPIO_PINS_PORT,GPIO_PINS_PIN_9_PIN | GPIO_PINS_PIN_11_PIN);
            break;

        case 4:
            DL_GPIO_clearPins(GPIO_PINS_PORT, GPIO_PINS_PIN_9_PIN | GPIO_PINS_PIN_7_PIN | GPIO_PINS_PIN_8_PIN | GPIO_PINS_PIN_2_PIN | GPIO_PINS_PIN_10_PIN | GPIO_PINS_PIN_11_PIN);
            DL_GPIO_setPins(GPIO_PINS_PORT,GPIO_PINS_PIN_6_PIN | GPIO_PINS_PIN_10_PIN | GPIO_PINS_PIN_11_PIN);
            break;

        case 5:
            DL_GPIO_clearPins(GPIO_PINS_PORT, GPIO_PINS_PIN_6_PIN |GPIO_PINS_PIN_10_PIN |GPIO_PINS_PIN_9_PIN | GPIO_PINS_PIN_7_PIN | GPIO_PINS_PIN_11_PIN);
            DL_GPIO_setPins(GPIO_PINS_PORT,GPIO_PINS_PIN_8_PIN | GPIO_PINS_PIN_11_PIN);
            break;

        case 6:
            DL_GPIO_clearPins(GPIO_PINS_PORT, GPIO_PINS_PIN_6_PIN |GPIO_PINS_PIN_10_PIN |GPIO_PINS_PIN_9_PIN | GPIO_PINS_PIN_7_PIN | GPIO_PINS_PIN_11_PIN);
            DL_GPIO_setPins(GPIO_PINS_PORT,GPIO_PINS_PIN_8_PIN );
            break;

        case 7:
            DL_GPIO_clearPins(GPIO_PINS_PORT, GPIO_PINS_PIN_8_PIN |GPIO_PINS_PIN_10_PIN | GPIO_PINS_PIN_7_PIN | GPIO_PINS_PIN_2_PIN );
            DL_GPIO_setPins(GPIO_PINS_PORT, GPIO_PINS_PIN_6_PIN |GPIO_PINS_PIN_9_PIN |GPIO_PINS_PIN_11_PIN|GPIO_PINS_PIN_12_PIN);
            break;

        case 8:
            DL_GPIO_clearPins(GPIO_PINS_PORT, GPIO_PINS_PIN_6_PIN |GPIO_PINS_PIN_9_PIN |GPIO_PINS_PIN_11_PIN|GPIO_PINS_PIN_8_PIN |GPIO_PINS_PIN_12_PIN);
            DL_GPIO_setPins(GPIO_PINS_PORT, GPIO_PINS_PIN_11_PIN);
            break;

        case 9:
            DL_GPIO_clearPins(GPIO_PINS_PORT, GPIO_PINS_PIN_6_PIN |GPIO_PINS_PIN_9_PIN |GPIO_PINS_PIN_8_PIN |GPIO_PINS_PIN_10_PIN | GPIO_PINS_PIN_11_PIN);
            DL_GPIO_setPins(GPIO_PINS_PORT, GPIO_PINS_PIN_11_PIN);
            break;

        default:
            break;
    }
}
```

(Here is a detailed logic of how to print numbers on the display.)

3.2. Hardware Development

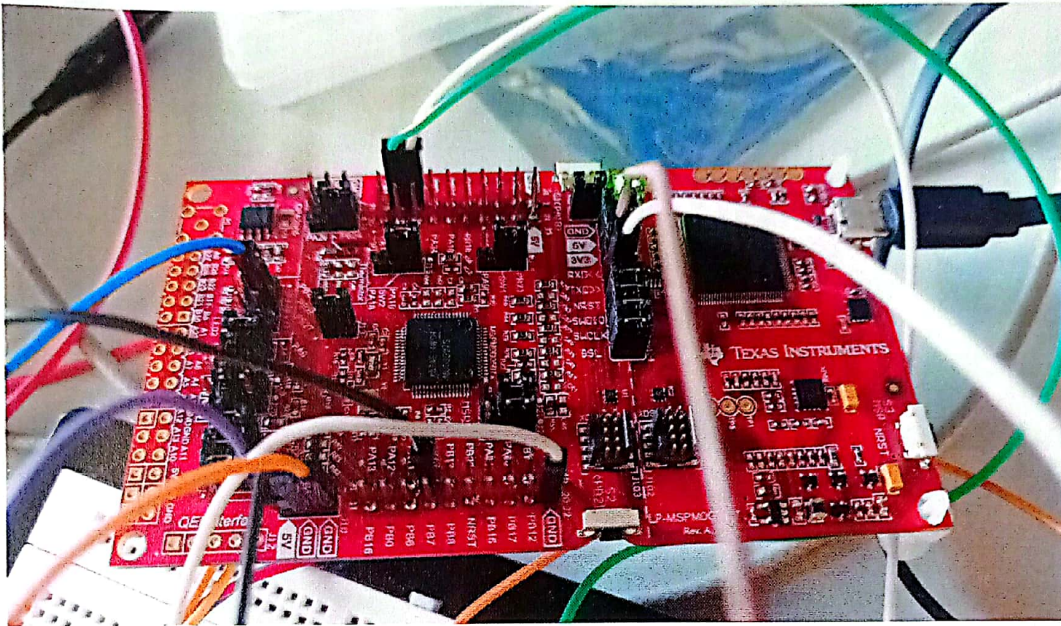
In this section I will cover the Hardware development.

I used two different microcontrollers in this project. First one is MSPM0G3507(red) and MSPM0C1104(green). I had to configure their pins both in coding and cables also. So, coding part has been explained and here is the cable part.

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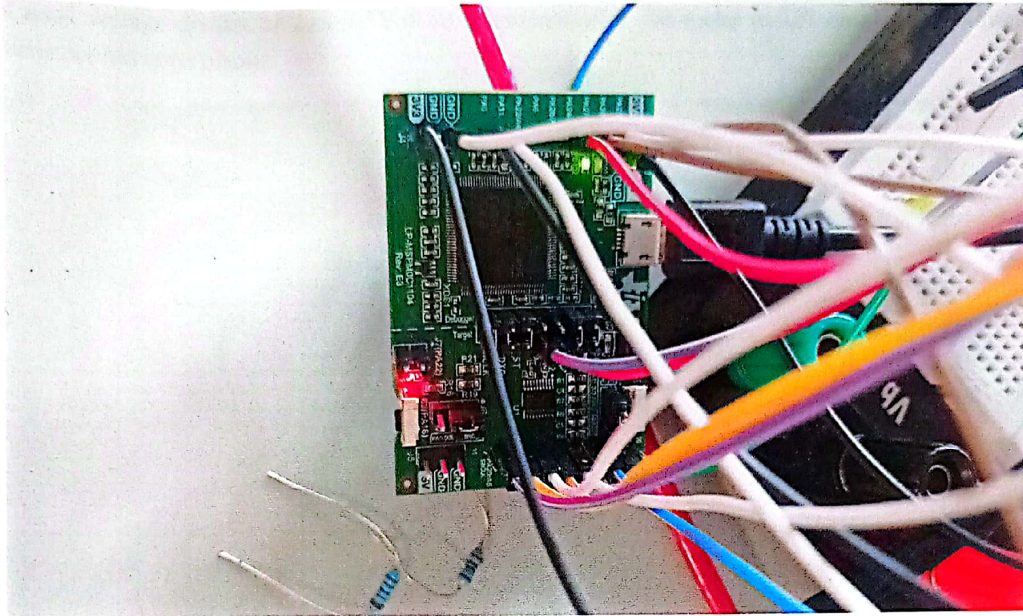

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Red microcontroller is used to get the measured distance value, so it had to be connected to ultrasonic sensor (HC-SR04) also to the computer to get energy. Their UART pins is connected to green microcontroller's UART pins to communicate them. Its PA27 and PB2 pins is connected to ultrasonic sensor's TRIG and ECHO pin respectively.



(The image of MSPM0G3507 microcontroller)

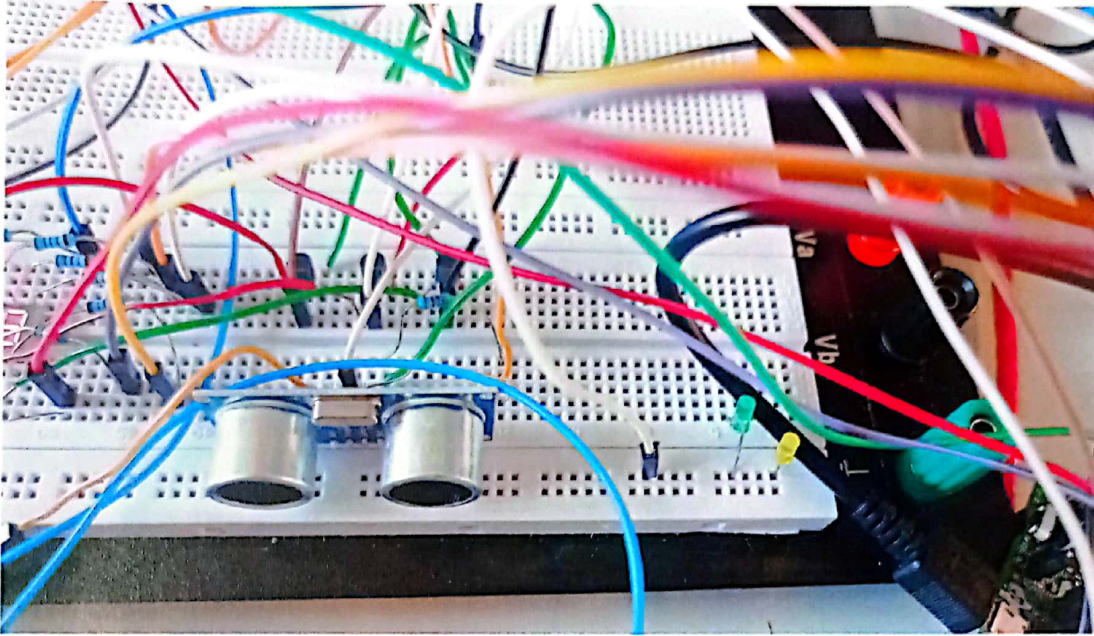
The green microcontroller is used to optimize the LED Display and it's 11 pins is connected to LED Display's 11 pin. As said 2 RX(Receiver) and TX(Transmitter) UART pins is connected to red microcontroller's ones.



(The Image of MSPM0C1104 microcontroller)

For the ultrasonic sensor (HC-SR04), 4 pins are connected as mentioned to red microcontroller. It works with 5 Volt, so VCC is connected to 5 Volt pin but to neglect to harm the microcontroller,

I used voltage divider to turn 3.3Volt to microcontroller. To make it 220 ohm and 470 ohm resistors has been placed.



(Image of Ultrasonic sensor(HC-SR04))

LED Display needed 11 pins to continue to work so 11 pins has been connected to green microcontroller. Because of working with LEDs, I had to insert resistors which are same and 1k

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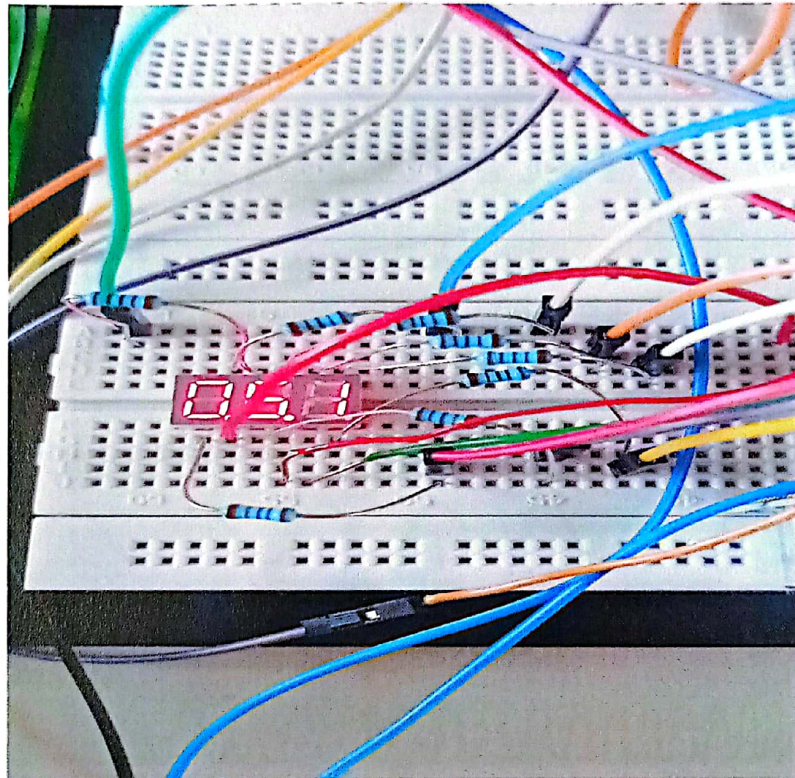
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ohm value to not to burn LEDs. I also used transistors to get a good view of result because without using as guessed resistors and LED consumes some energy and it decreases the power of light of LEDs.



(Image of LED Display)

3.3. Sensor Experiment

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3.3.1 Experiment of sensor curve

As mentioned in the first days, I was having an experiment from the company itself without relation my project. It was an experiment to measure the curve of the distance measurement under different degree conditions like 0°, 40°, 60° etc. It gave me a good insight into sensor technology. They showed me how to save the data, how to shaft works, how sensor sends the data to computer and so on. It was a fun and educated experience of mine.

SOUND WAVE

Infrasound
(below 16 Hz)



Audible frequencies
(16 Hz - 20kHz)



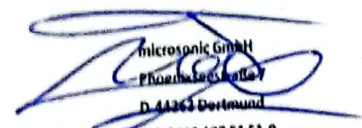
Ultrasound
(over 20 kHz)



(Here is the illustration of Ultrasonic Wave)

3.3.2 Experiment of ultrasonic sensor of my project

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Marmara University Training Report

In my project it is important to see the results of every step or every debugging. I use waves with helping of clock of TIMER of microcontrollers, so I had to see if they are working correctly or not. So, I used a program called as Logic 2 to see the desired waves and their movements. The program is so useful to see and analyze. TRIG and ECHO waves of ultrasonic sensor and UART waves (RX and TX) of microcontrollers can be seen so easily. These are the reasons I used that program.



(Here is image of the Logic Program)

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4. CONCLUSIONS

As a conclusion I have learnt too many things in this Internship Training. I had great experiences and met great people in the work area. I practiced my old knowledge about Programming in Embedded Systems and Sensors. Using Code Composer Studio and C language gives a lot of easiness to the Programmer side. It has different techniques, huge amounts of libraries to dive into, brief usage and so on. So personally, I liked to work with C language and such a great atmosphere. Therefore, I enjoyed coding, studying and working in the company.

I had taken C, C++, MATLAB and Java lectures in my own school, so it helped me a lot during the Internship. This project helped me to criticize my future leaning.

C is an old and advanced language. It grows every day. Firstly, I was educated to gain sights about sensor and microcontrollers. It was a successful Internship. I criticized myself and my old knowledge about coding, so it is improved. I am so happy to do my internship in Microsonic GmbH. I did not make it to work with LED Display when I was preparing a project in Marmara University because of lack of time and experience but if I have a chance to make another similar project to this I am confident and feel to challenge and make accomplished.

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